

EUGENE L. BUCHANAN

Technical Proposal — Thermal Platform Extension

Apple Valley, CA · +1 (909) 545 5384 · eugene@serviceofothers.org · github.com/eugeneb50



September 26, 2026

Exowatt — Founders, Engineering, and Commercial

Miami, FL

International deployment — community energy hubs in developing nations

Attached: follow-up to Exowatt Talent Network application

Re: The P3 platform as a community energy hub — agricultural residue to biochar, heat, and power, built on one shared thermal bus

Dear Exowatt team,

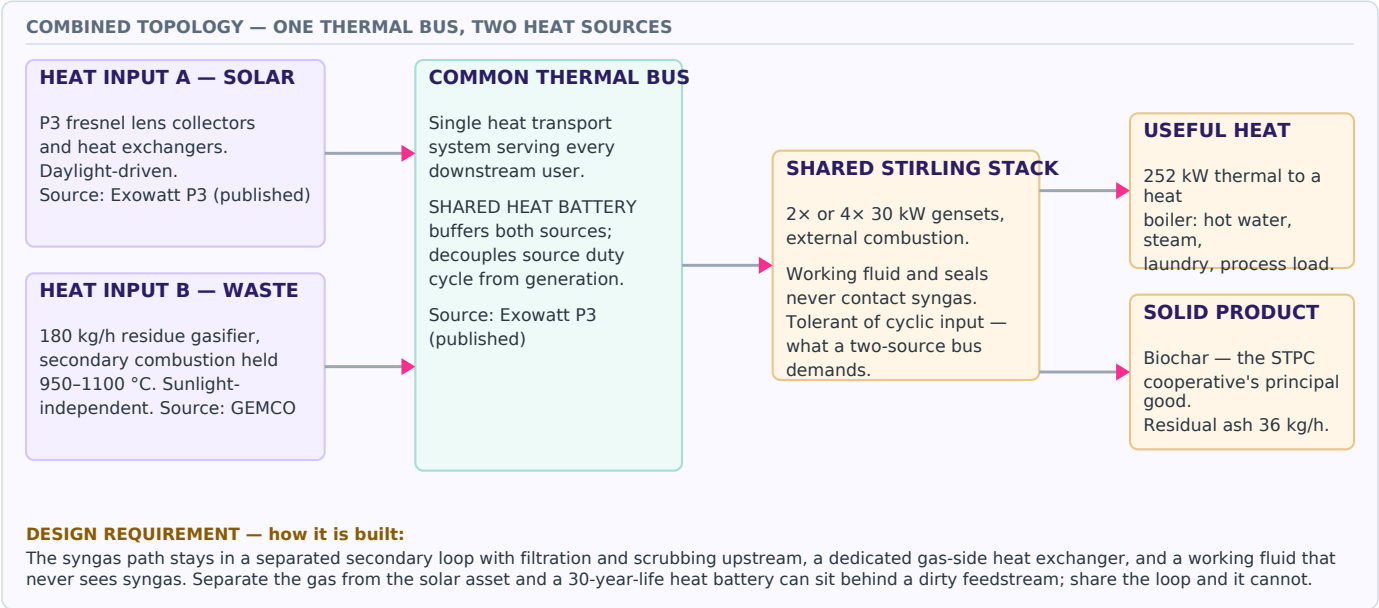
The P3 platform concentrates solar heat with proprietary fresnel lenses and heat exchangers, stores it in a heat battery, and dispatches it through your own heat engine on demand. That is a capture-store-dispatch architecture, and those three functions are also the three functions of a different product. There is a second market sitting on the hardware you have already built, and it runs on the same bus: a gasification unit charging your heat battery and feeding your Stirling stack, so the plant keeps dispatching when the sun is not out. I would like to work on it.

I am not proposing that Exowatt change strategy. Data centers pay for the R&D, the manufacturing scale, and the 30-year service life. I am proposing a second application where the feedstock is a liability somebody else wants removed and the heat value is consumed locally rather than exported to a grid. That is a deployment problem, not a technology development problem, and it is the part I can carry.

Why I am the person to work on this

The role inventory below is not a capability list. It is the set of functions this project needs, each matched to something I have already done. Everything in it is drawn from my own record.

Function this project needs	What I have actually done
Thermal and energy engineering	Twenty-two years in solar and waste-to-energy. More than \$32M in solar, wind, and geothermal installations sold and integrated. Air conditioning estimation, air flow and BTU calculation, envelope analysis, seasonal thermal load.
Test and reliability programs	Built a test automation function from scratch. WHQL certification testing of high-availability clusters under sustained load with injected drive failure. WHQL-quality evidence programs, 30 years of test ownership.
Permitting and entitlement	Permit-ready CAD packages — plot plans, single-line electrical, string impedance, NEC 690 safety calcs, rack load and footing calcs — through licensed-engineer stamping, city plan review, sign-off, and permit to operate. Recovered a non-permitted PV installation into compliance.
Public governance and board process	Four years as an elected California public-agency president. Brown Act and open-meetings compliance through active litigation; board handbook and policy revisions; hiring and replacing executive staff.
Capital and cost control	Cut operating expenditures 53% on a \$16,027,777 annual budget while protecting capital projects. Tax credit applications, grant and licence filings at Federal, State, and City level. Insurance and risk-coverage delegate.
Community and social enterprise	Built a nonprofit and community outreach program from nothing, including the Federal, State, and City licensing needed to stand it up. Public speaking and community events in front of residents and staff.
International operations	Chief Technology Officer of a Korean solar marketing company — sales, procurement, permitting, and finance across an international operation.
Software, telemetry, and data	Author of a 51-tool Rust Model Context Protocol server, 299 tests passing. RAG assistant shipped under SOC 2 controls. Satellite prospecting pipeline using Sentinel-2 spectral screening, SRTM terrain analysis, and GIS field data.



Who supplies what, and what is already published

Function in the combined system	Contributor	Published figure relied on
Heat input A — solar	Exowatt P3	Proprietary fresnel lens collectors and heat exchangers
Heat input B — waste, sunlight-independent	GEMCO HG-120	180 kg/h feedstock; 3-5 cm at ≤25% moisture; 950-1100 °C secondary combustion
Heat transport	Combined — new	One common bus to every downstream user
Thermal storage	Exowatt P3 — shared	Heat battery; extended storage with low losses
Power conversion	GEMCO — shared stack	2x or 4x 30 kW external-combustion Stirling gensets
Useful-heat tap	GEMCO + STPC	252 kW thermal (HG-120) to hot water, steam, laundry, distilled water
Solid product	STPC + GEMCO	Biochar; 36 kg/h residual ash, stated usable as construction material
Site output	Combined	Electricity, useful heat, and biochar, off-grid capable; 1-2 × 40 ft modular; 15-20 yr gasifier life

All figures as published by the named source. GEMCO models are HG-60 and HG-120; the HG-120 is quoted. P3 details from Exowatt's published product description. STPC details from the cooperative's published programme. Nothing above is a Exowatt commitment — it is a map of what exists.

Where the integration is real

One bus, two heat sources. The gasification plant does not replace your power train — it feeds the same heat storage and the same Stirling generator stack your solar concentrators feed. Solar charges the battery by day; the gasifier charges it at night, in cloud, and in winter. Solar concentration is the cheaper, cleaner source and carries the base load, but it is the source you cannot schedule. A second, sunlight-independent source on the same bus turns an intermittent plant into a dispatchable one, running on a fuel stream that is a liability somebody is already paying to remove.

Stirling is the right machine on a bus like this, and I would keep it. GEMCO's gensets are external-combustion, so the working fluid and the seals never contact syngas. Stirlings also tolerate cyclic heat input, which is exactly what buffering two sources with different duty cycles demands. A single prime mover sized for one source is a compromise; a shared stack fed from a buffered bus is not.

Feedstock arrives presorted and dried. Sorted by material and moisture, and dried to the gasifier specification before it reaches the feed hopper. That is a specification, not a topic — it is how the unit is fed, and the rest of the system is designed on the assumption that it has already been met.

Keeping the gas path separate is what makes the shared bus work. Syngas carries sulfur, tars, and particulates, and a 30-year-life solar asset should never meet them directly. So the design separates them: a dedicated gas-side heat exchanger, particulate filtration and scrubbing upstream, and a working fluid that never contacts syngas. At Desert Power I did process stabilization, hot gas flow control, and anti-fouling research for Fischer-Tropsch syngas — the same contamination problem in a different reactor. That experience is what makes this a solvable design problem instead of an open risk, and it is why the thermal margin on the solar side stays where your warranty put it.

What the unit actually is

The unit is a community hub, not a power plant with a customer. Every output lands on something somebody in the village uses every day: electricity for the cooperative's processing and lighting, hot water and laundry through the useful-heat tap, clean water, and biochar as a soil amendment and a cooking-fuel substitute that takes pressure off the surrounding woodland. Nothing on that list is an export. The value is created on site and consumed on site, which is what makes the economics work at a scale too small to interest a utility.

Connectivity is part of the same system, not a separate problem. A hub this disruptive to daily life needs a communications link to run on — remote monitoring, dispatch decisions, payments for biochar and surplus energy, and the telemetry that makes a 15–20 year asset serviceable instead of a mystery. LEO satellite backhaul with local WiFi node stations gives the hub a link that does not depend on a grid that was never going to reach it, and it lets a fleet of hubs be monitored and dispatched as one system from anywhere in the world. Connectivity is where a remote deployment either becomes operable or becomes a photograph.

The fuel is local, so the supply chain can be too. The published GEMCO configuration is modular and containerised by design, and that modularity is exactly what makes local manufacture practical — vessels, frames, heat exchangers, skids, and control cabinets can be built and serviced in the regions where the units deploy, with no import logistics, no tariff exposure, and a spare-parts supply chain that is not an ocean away from the technician. For a product whose entire purpose is local resilience, depending on a single international manufacturing origin is a liability, and a distributed one is an advantage. The first deployment should be assembled locally with locally made parts, and that should be the standard, not the exception.

Ecologically, the arithmetic is favourable and the numbers should be published. Agricultural residue that is currently burned in the field is removed from the equation, and open-burning of crop residue is a recognised seasonal air-quality and health problem across the regions this would serve. Biochar returned to agricultural soil holds carbon and improves water retention on exactly the land that produced the feedstock. This is a measurable, verifiable improvement rather than a claim, and I would want the carbon accounting done properly and published, because that is what makes it financeable and what keeps it honest.

Why the second market is strategy, not diversion

Data center revenue is concentrated, and concentration is a limit. A handful of very large buyers produce very large contracts — and they also produce a reputational position that is contested in the communities those facilities are being sited in. That contention is real and it is a permitting and community-relations exposure that no amount of good engineering removes.

Distributed municipal and community power is diffuse, and diffusion is a funnel. Many small buyers, many small communities, and a revenue base with no single point of failure. It opens channels a data center sale does not — distributors, development-finance capital, municipal and cooperative procurement — and reference installations are reusable in a way that one hyperscale win is not. Every hub that goes in makes the next one easier to permit, to finance, and to sell.

The social enterprise frame is the differentiator, and it is also the most durable thing here. STPC is a community cooperative serving more than 5,000 seasonal agricultural workers in the San Quintín Valley — biochar, clean water, hot showers, and a community laundry on infrastructure the community operates. That is not a marketing angle bolted onto a technology. It is the reason the technology is permitted, staffed, and politically durable where a private operator would struggle to site at all. For an investor base that includes climate-focused funds, a demonstrable operating asset serving a migrant-worker valley survives scrutiny in a way a purchased carbon offset increasingly does not.

The environmental stance improves, it does not get a paragraph bolted on. A company that powers AI data centers has a real and increasingly awkward question to answer about what its hardware does to the places it is built. A platform that also converts a local waste liability into heat, power, water, and soil amendment — and that does it in the developing-world communities those questions are usually asked about — has an answer that is demonstrable rather than asserted. That is worth having before anyone asks. It is worth more after they ask.

How this gets funded, and my part in it

What makes me useful here is that I am not a specialist. A project like this fails at the seams: the heat exchanger gets designed and nobody can permit it; the cooperative is formed and nobody can raise the money; the unit is funded and nobody can keep it running for fifteen years. Those are the seams I have spent thirty years working, and they are why my record is broad rather than deep. A company of specialists delivers the heat exchanger. This project needs somebody who can get a permit stamped in one jurisdiction, restructure a procurement chain in another, and sit down with a cooperative board in a third.

I can work the capital stack, and I would rather it be sequenced than rushed. This kind of project has three distinct funding populations, and they are not interchangeable. Early-stage feasibility and a small pilot come from impact and catalytic capital, corporate social responsibility, diaspora and development-finance money, where

the ticket is small and the diligence is light. Commercial scale comes from carbon and biochar offtake plus equipment debt — the biochar is the line that makes it bankable, because a verified carbon drawdown on a real waste stream is a product with a market, not a projection. And institutional money — Green Climate Fund, Global Environment Facility, World Bank DevelopEx, the bilateral windows — is the tranche that makes a portfolio possible. Each has different diligence, different timelines, and different paperwork, and most of these projects die between tranches rather than inside any one of them.

The honest sequence, which is why I am proposing Stage 0 first. Small money gets a measured pilot. Measured pilots get the accredited-finance track open, as a co-application with an entity that holds the accreditation. Of the four tranches, only the first two are things I would ask for before there is operating data, and both of them are within reach of a company of your size and mine. I would not bring you a GCF application and ask you to wait two years for a first disbursement.

Tranche	Capital source	What it funds	My part
Stage 0-1	Founder capital, CSR, or a feasibility grant	Desktop study, feedstock survey, bench gasifier, first stakeholder sign-offs.	Build the business case, mass and energy balance, and a funder-ready dossier.
Stage 2	Impact and catalytic capital, diaspora and diaspora-adjacent investment, development finance, CSR partnerships	One container-class pilot: gasifier on the same bus as the solar array, community heat, water and laundry load, LEO backhaul and WiFi node stations.	Structure the deal, write the investor materials, run co-application, hold the permitting schedule.
Stage 3	Carbon and biochar offtake, plus equipment debt	Replication of the hub into additional communities.	Measurement, reporting and verification, offtake contracting, unit sequencing, local manufacture ramp.
Stage 4	Blended finance — GCF, GEF, World Bank DevelopEx, bilateral windows, via an accredited entity	Portfolio scale across regions.	Partner selection, accreditation, project documents, environmental and social impact assessment, and the reporting that keeps the money flowing.

The one structural fact worth stating plainly: GCF and GEF capital is accessible to accredited entities, not to a small company directly. So the realistic sequence is a small first pilot funded by impact capital, with the accredited-finance track opened as a co-application once a hub has measured data behind it. Companies that go to GCF first with nothing operating generally do not get funded. The order matters more than the ambition.

What I would want to validate, in order

Stage 0 — desktop	Pick one target region and one partner community. Feedstock audit: volume, material mix, seasonality. Energy and mass balance. Connectivity survey. Permitting and local-content review. No capital.
Stage 1 — bench	Gasifier on a bench scale, measured. Does the syngas drive the Stirling, and at what efficiency? Does the char come out to spec? Most important: fouling rate on a heat exchanger across weeks rather than days, and whether the separated gas loop holds the working fluid clean.
Stage 2 — pilot	One container-class gasifier on the same bus as the solar array, feeding a community heat load, a water and laundry load, and a measured electrical load through the shared Stirling stack. LEO backhaul and WiFi node stations commissioned with the monitoring plan, not after it. Full permitting package, stamped and through plan review.
Stage 3 — cooperative ownership	Local manufacture of vessels, frames, heat exchangers, skids, and control cabinets. Transfer operations, maintenance, and revenue share to the partner cooperative. Where value is created locally, and where the second, third, and fourth hub stop being an ask.

What I am proposing is not a role on your team. It is a function you probably do not have: somebody who can take this from a published reference design to a permitted, financed, monitored, cooperatively operated hub in a country neither of us has worked in, and who can raise the money for the next one while the first is still running. Waste out of the environment, power and heat and water and soil amendment into a community, a cooperative holding the asset, a supply chain that is local because the product is. The environmental story does not need to be written for this platform — it gets built in. I would start at Stage 0, because it costs nothing and it is where an honest answer comes out. I am ready to relocate to Austin or Miami, and I would like to start on this.

Best regards,

Eugene L. Buchanan

Quality, Test & Controls Engineer · Relocating to Austin, TX or Miami, FL
eugene@serviceofothers.org · +1 (909) 545 5384 · github.com/eugeneb50
STPC · stpc@encuentralo.mx